



Introduction to Artificial Intelligence (AI)

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What is AI?

- Artificial Intelligence is an attempt to make a computer, a robot, or other piece of technology 'think' and process data in the same way as we humans do.
- AI is a branch of science which deals with helping machines find solutions to complex problems in a more human-like fashion.
- AI therefore has to study how the human brain 'thinks', learns, and makes decisions when it tries to solve problems or execute a task.
- The aim of AI is to improve technology by adding functionality related to the human acts of reasoning, learning, and problem-solving.
- Example : Home Automation Systems, Cortana is example of a voice controlled intelligent system



Need of AI

- The need for AI is fueled by the fact that it is a technology that can enhance machines by equipping them with intelligence.
- The technology is used to have machines help humans by teaching themselves to adjust, adapt, refer to more data and process that quickly in order to provide a for a better or alternative answer where possible.
- AI is to create technology that allows machines to function in an intelligent manner with or without human supervision
- With the help of AI, you can create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
- With the help of AI, you can create your personal virtual Assistant, such as Alexa, Google Assistant, Siri, etc.
- With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.



Need of AI

- Makes Search faster and simpler
- Makes interaction smarter
- Provides Relevant customer experience
- Offers personalized and custom e-commerce experience
- Provides effective means of carrying out business.



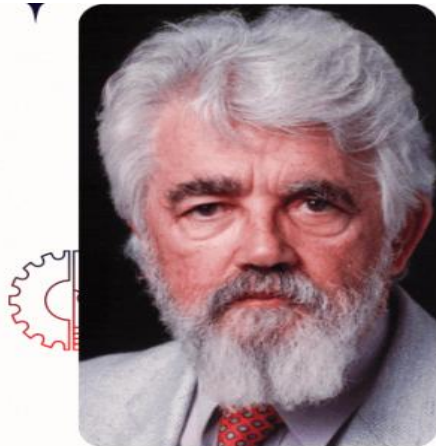
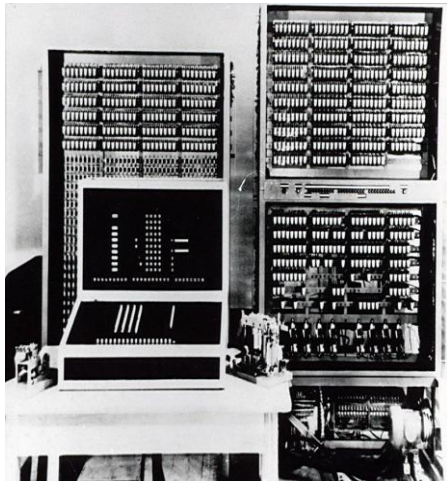
History of AI

1950's	 Artificial intelligence (AI) <i>Human intelligence exhibited by machines</i>		
	1980's	 Machine learning <i>AI systems that learn from historical data</i>	
	2010's	 Deep learning <i>Machine learning models that mimic human brain function</i>	
	2020's	 Generative AI (Gen AI) <i>Deep learning models (foundation models) that create original content</i>	

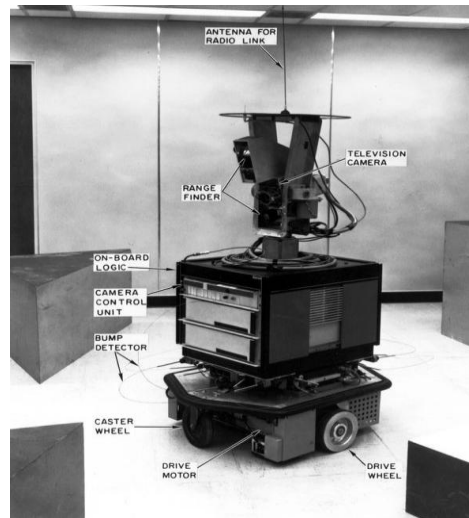


History of AI

- 1941: First electronic Computer
- 1956 : John McCarthy coined the term 'artificial intelligence' and had the first AI conference.
- 1969 : Shakey was the first general-purpose mobile robot built. It is now able to do things with a purpose vs. just a list of instructions.
- 1997 : Supercomputer 'Deep Blue' was designed, and it defeated the world champion chess player (Garry Kasparov) in a match. It was a massive milestone by IBM .



John McCarthy
Father of artificial intelligence



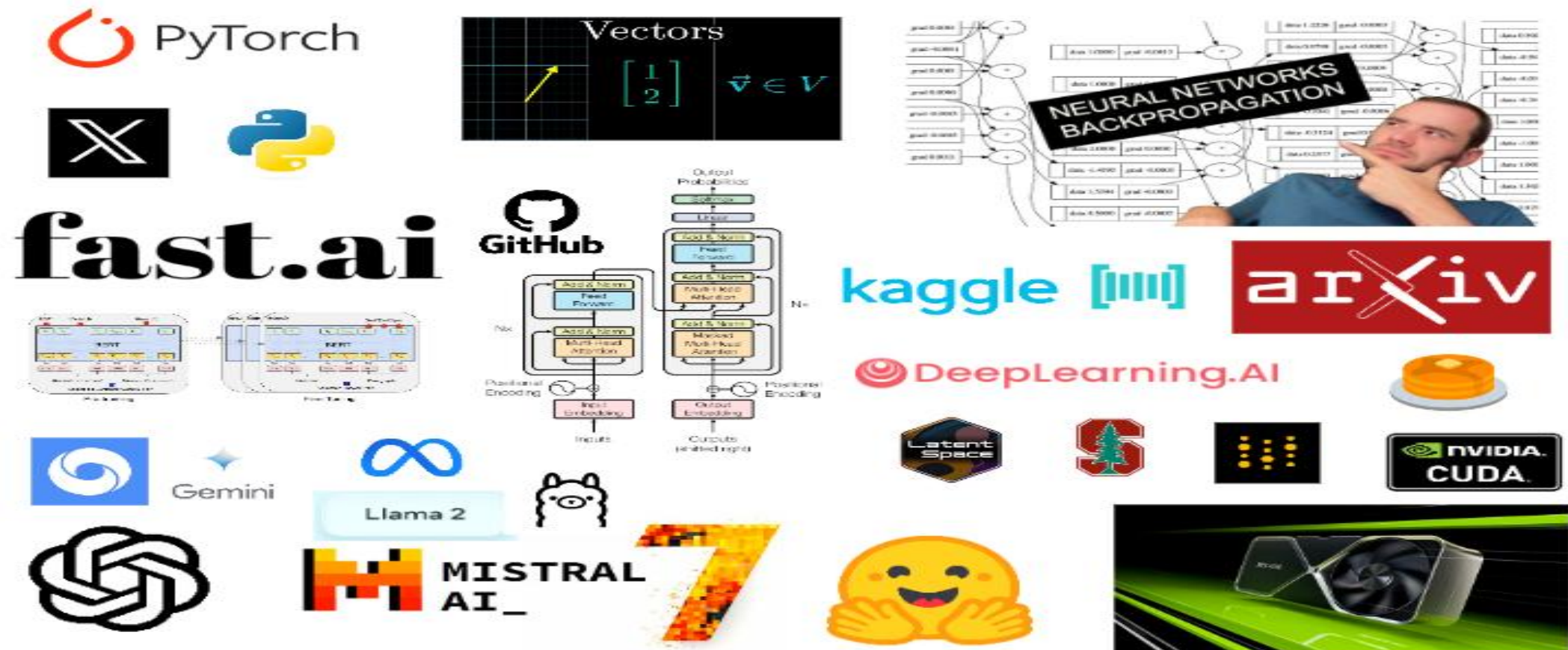
History of AI

- 2002 : The first commercially successful robotic vacuum cleaner was created.
- 2005 - 2019 : Today, we have speech recognition, robotic process automation (RPA), a dancing robot, smart homes, and other innovations make their debut.
- 2020 : Baidu (Chinese multinational technology company) releases the LinearFold AI algorithm to developed a vaccine during the early stages of the SARS-CoV-2 (COVID-19) pandemic. The algorithm can predict the RNA sequence of the virus in only 27 seconds, which is 120 times faster than other methods.



History of AI

- 2020-2024 : Generative AI (gen AI) uses combination of text, voice, images, and video to offer more comprehensive user interactions.



Generative AI

- **ChatGPT or DALL-E:** Generative artificial intelligence created by OpenAI, a Microsoft-backed, profit-capped company with the mission to develop artificial intelligence to serve humankind
- **Google Bard(Gemini):** Google's generative AI with integrations to Google products like Google Lens and Gmail.
- Health care, manufacturing, software development, financial services, media and entertainment, and advertising and marketing.

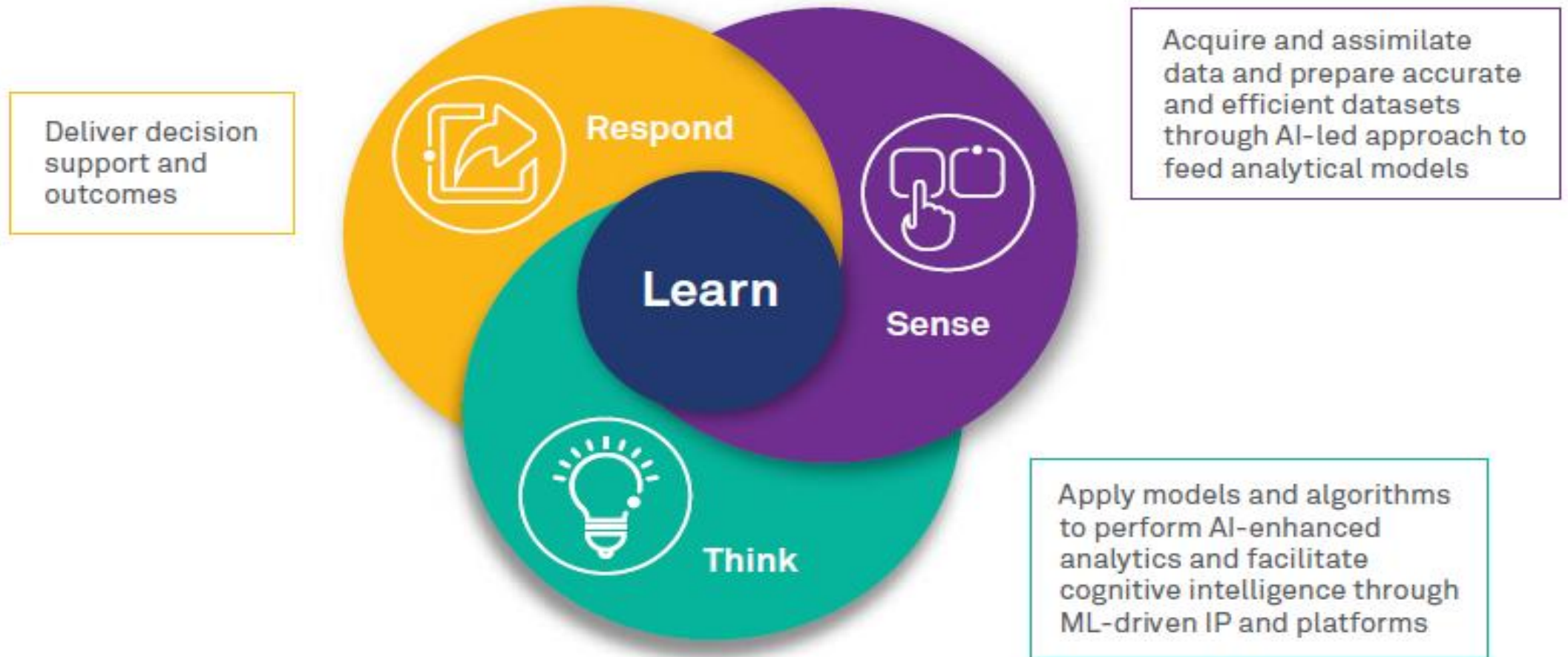


AI Understanding

- Help machines find solutions to complex problems in a similar way that humans do and apply similar logic in the form of heuristic (self-learning) algorithms to calculate and display the required output.
- Intelligence can be loosely defined as the capability to obtain knowledge and skills and to apply those to various situations without supervision.
- **Intelligence is made up of following also known as AI Programming Cognitive Skills**
 - Learning
 - Reasoning
 - Problem Solving
 - Perception
 - Linguistic Intelligence



AI Understanding



AI Programming Cognitive Skills or Components of AI

Learning

- Learning for AI includes the trial-and-error method.
- It includes memorizing individual items like different solutions to problems,

Reasoning

- It is to allow the platform to draw inferences that fit with the provided situation

Problem-solving

- AI's problem-solving ability comprises data, where the solution needs to find the features.

Perception

- the element scans any given environment by using different sense-organs, either artificial or real.

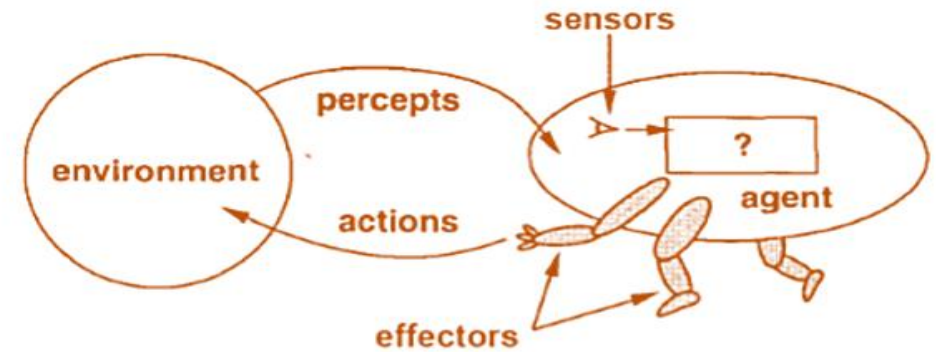
Language-understanding

- distinctive types of language over different forms of natural meaning



AI Elements : An Agent

- AI system is composed of two main elements :
 1. Agent and
 2. Its surrounding Environment
- An AI agent is a system that is –
- Perceiving its environment through **sensors** and
- Acting upon that environment through **actuators/effectors**
- The intelligence of agents is calculated by their ability to create goals and achieve them.
- Anything' that can gather information about its environment and take action based on that information.
- An agent can be:
 - **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
 - **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
 - **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

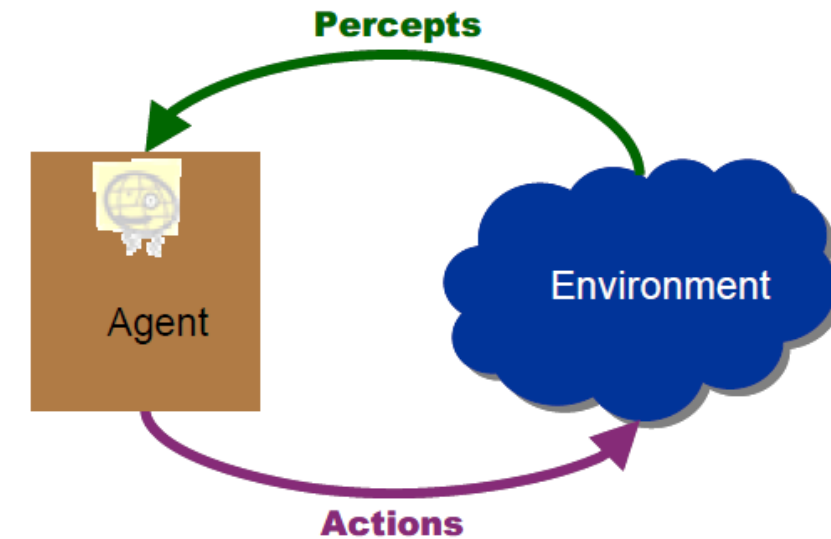


AI Elements : Environment

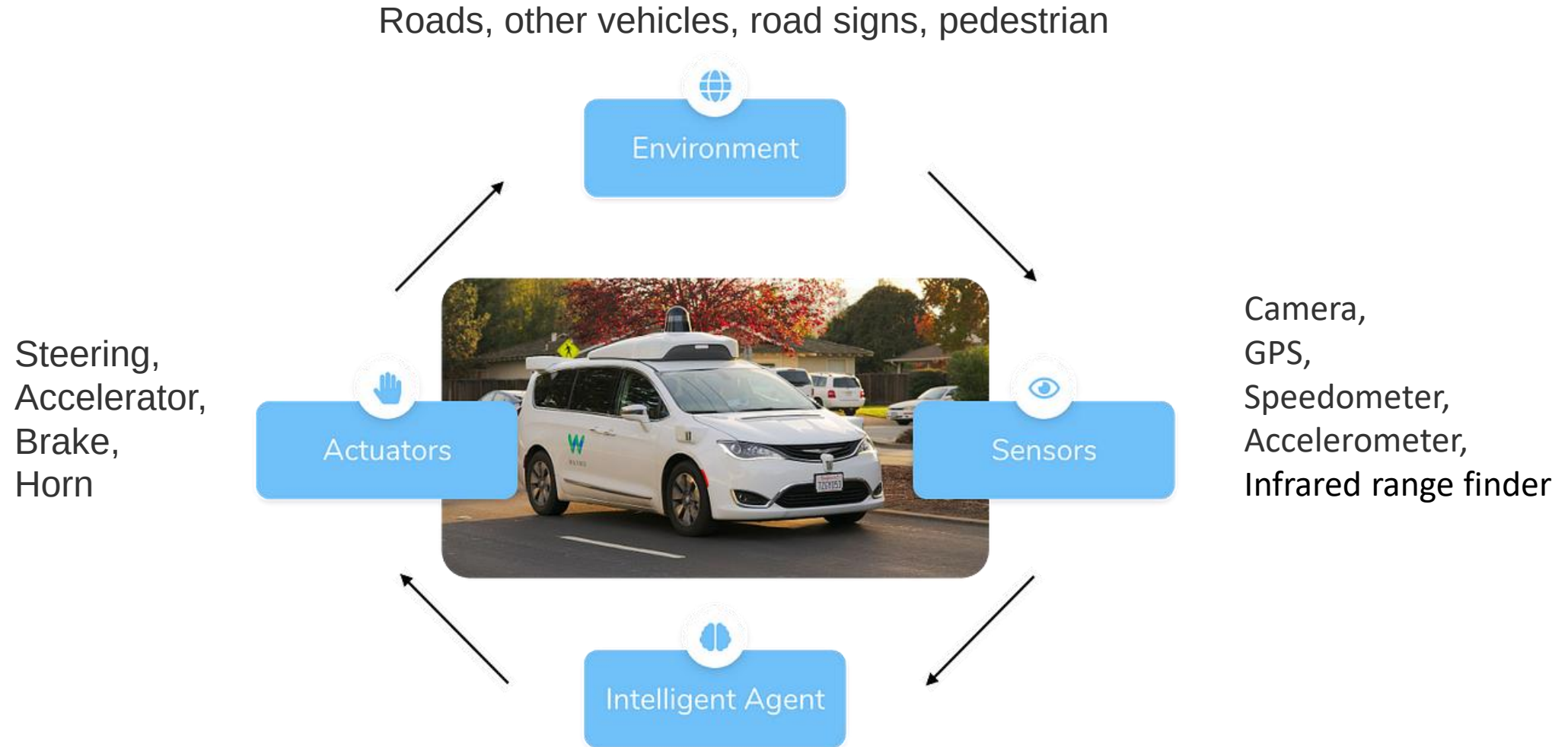
- An environment in artificial intelligence is the surrounding of the agent.
- The agent takes input from the environment through sensors and delivers the output to the environment through actuators.

Artificial Intelligence — Agents and Environments

Agents		Environments
Robot	→	Room
Chatbot	→	Chatting
Vehicle	→	Road
Program	→	Data & Rules
Machine	→	Working Field



AGENT & ENVIRONMENT



Types of AI

1. Reactive machines/ Purely Reactive

2. Limited memory

3. Theory of mind

4. Self-awareness



Types of AI : Four main types of AI

Reactive machines/ Purely Reactive

- Basic form of AI.
- They operate purely based on the present data and do not store any previous experiences or learn from past actions.
- These systems respond to specific inputs with fixed outputs and are unable to adapt.
- **Examples:**
- **IBM's Deep Blue**, which defeated the world chess champion Garry Kasparov in 1997. It could identify the pieces on the board and make predictions but could not store any memories or learn from past games.
- **Google's AlphaGo**, which played the board game Go using a similar approach of pattern recognition without learning from previous games.



Types of AI : Four main types of AI

Limited Memory

- Can learn from past data to improve future responses.
- Most modern AI applications fall under this category.
- These systems use historical data to make decisions and predictions but do not have long-term memory.
- Machine learning models, autonomous systems and robotics, often rely on limited memory to perform better.
- **Examples:**
- **Self-driving cars:** They observe the road, traffic signs, and movement of nearby cars, and make decisions based on past experiences and current conditions.
- **Chatbots** that can remember recent conversations to improve the flow and relevance of replies.



Types of AI : Four main types of AI

Theory of Mind

- Aims to understand human emotions, beliefs, intentions, and desires as well as interact socially.
- While this type of AI remains in development, it would allow machines to engage in more sophisticated interactions by perceiving emotions and adjusting behavior accordingly.
- . However, a machine based on this type is yet to be built.
- **Potential Applications:**
- **Human-robot interaction** where AI could detect emotions and adjust its responses to empathize with humans.
- **Collaborative robots** that work alongside humans in fields like healthcare, adapting their tasks based on the needs of the patients.
- Avengers, Robot, Transformer, Ra One, etc



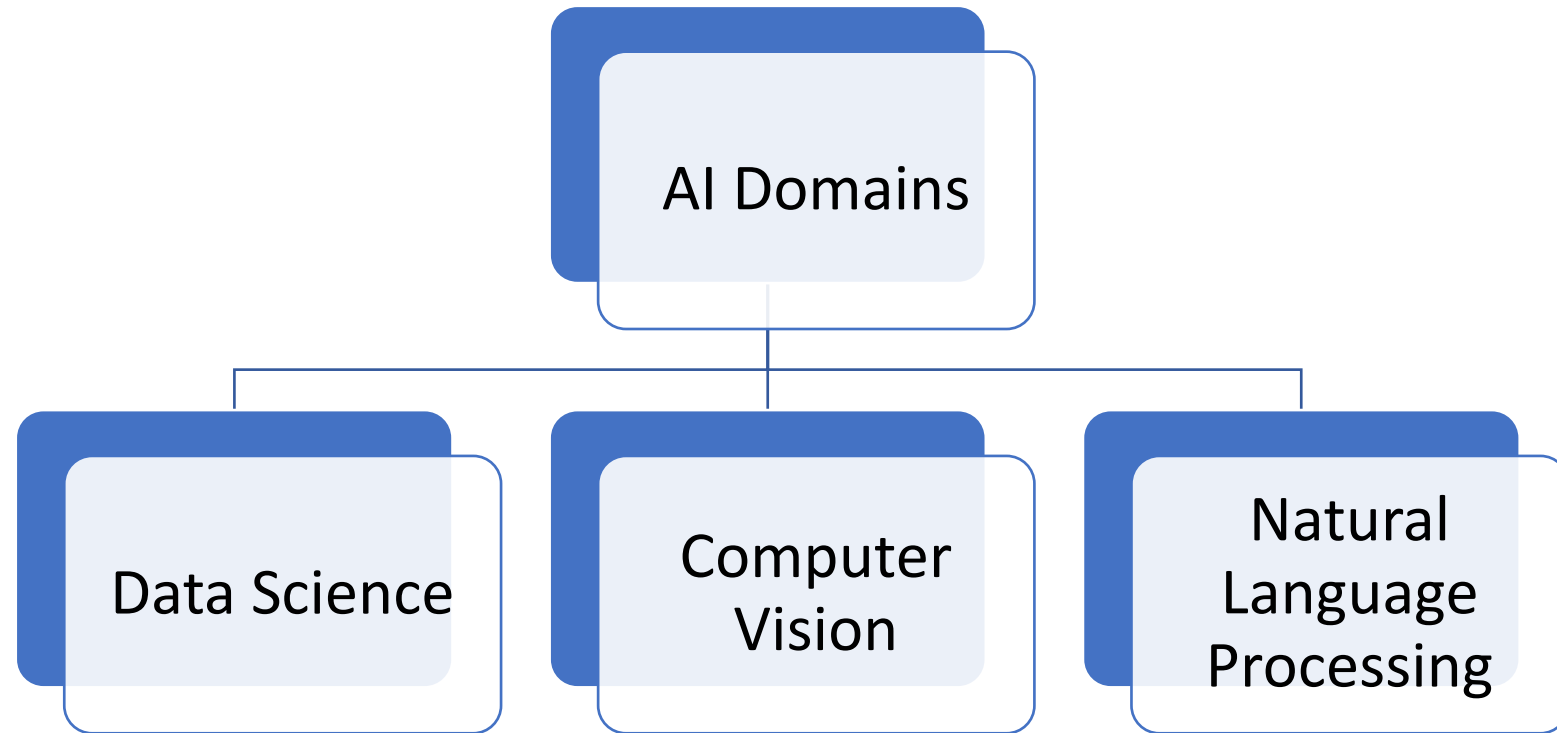
Types of AI : Four main types of AI

Self Awareness

- An advanced stage of AI that possesses self-consciousness and awareness.
 - This type of AI would have the ability to not only understand and react to emotions but also have its own consciousness, similar to human awareness.
 - While we are far from achieving self-aware AI, it remains the ultimate goal for AI development if they ever reach this level.
-
- **Potential Applications:**
 - Fully autonomous systems that can make moral and ethical decisions.
 - AI systems that can independently pursue goals based on their understanding of the world around them.



Main Domains of AI Technology



1. What is Data Science ?

- Data science is an inter-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from many structural and unstructured data.
- Data science is related to data mining, machine learning and big data
- It processes the numeric /alphanumeric data with data systems to extract information to make decision or predict the outcome for the problem statement.
- Example:

1. Recommendations



2. Price Predicting



3. Weather Forecasting



Artificial Intelligence:

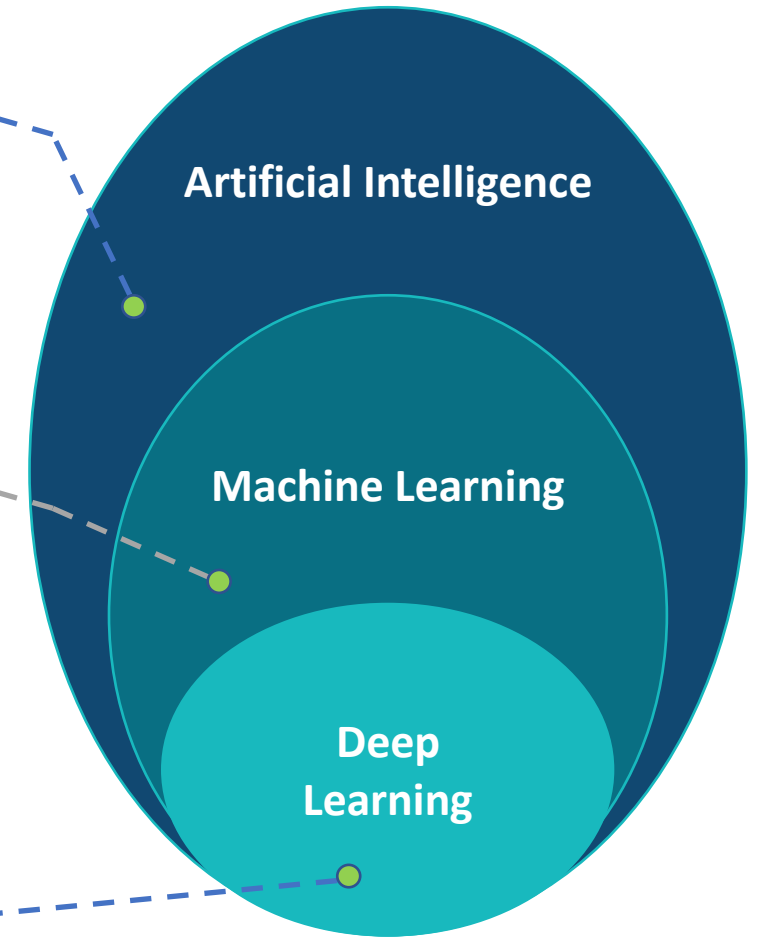
- A technique which enables machine to mimic human behavior

Machine Learning:

- Subset of AI which uses statistical methods to enable machines to improve the experience

Deep Learning:

- Subset of ML which makes the computation of multi-layer neural network feasible



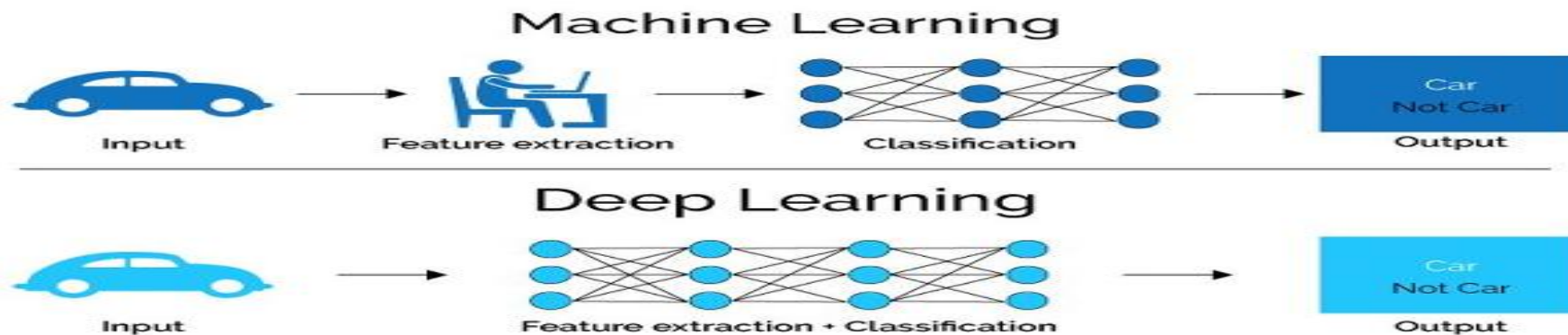
Ways to implement AI

Machine Learning

- It is machine learning that gives AI the ability to learn.
- This is done by using algorithms to discover patterns and generate insights from the data they are exposed to

Deep Learning

- Deep learning, which is a subcategory of machine learning, provides AI with the ability to mimic a human brain's neural network.
- It can make sense of patterns, noise, and sources of confusion in the data.



What is machine learning ?

- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.
- Machine Learning is the science (and art) of programming computers so they can learn from data.
- **Machine Learning (ML):** Involves training algorithms to recognize patterns in data and make decisions or predictions without being explicitly programmed.
- **Deep Learning(DL) :** A subset of machine learning that uses neural networks with many layers to analyze large amounts of data for highly accurate predictions.



Examples of ML Applications

- Analysing images of products on a production line to automatically classify them
 - This is image classification, typically performed using convolutional neural networks
- Detecting tumors in brain scans
 - This is semantic segmentation, where each pixel in the image is classified (typically use CNNs)
- Automatically classifying news articles
 - This is natural language processing (NLP), and more specifically text classification
- Automatically flagging offensive comments on discussion forums
 - This is also text classification, using the same NLP tools
- Forecasting your company's revenue next year, based on many performance metrics
 - This is a regression task (i.e., predicting values) that may be tackled using any regression model



Examples of ML Applications

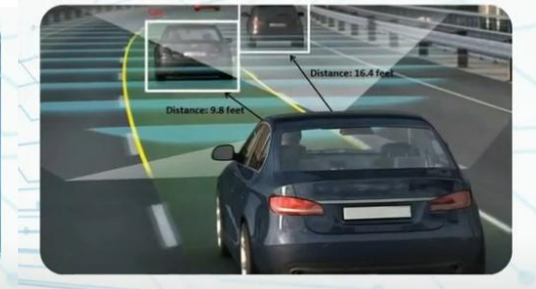
- Detecting credit card fraud
 - This is anomaly detection example
- Segmenting clients based on their purchases so that you can design a different marketing strategy for each segment
 - This is clustering example
- Representing a complex, high-dimensional dataset in a clear and insightful diagram
 - This is data visualization, often involving dimensionality reduction techniques
- Recommending a product that a client may be interested in, based on past purchases
 - This is a recommender system



2. Computer Vision

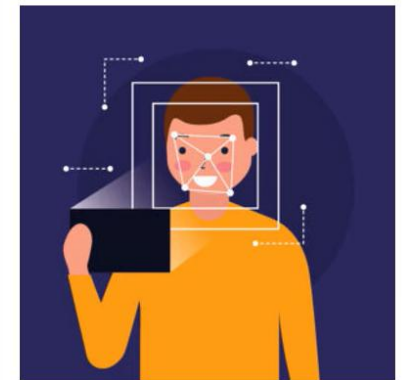
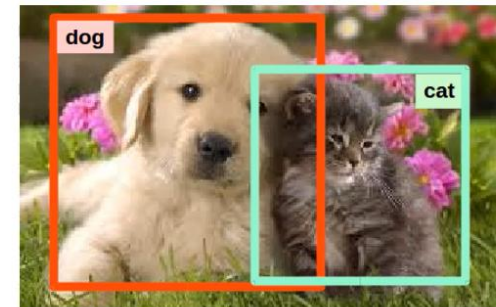
- Computer vision focuses on enabling machines to interpret and make decisions based on visual input, such as images or videos.
- .Computers can use deep learning techniques to comprehend images in the same way that humans do.
- Computer vision has several applications, such as the following:
 - Content moderation to automatically remove unsafe or inappropriate content from image and video archives
 - Facial recognition
 - Image classification

Does many calculations in order to recognize patterns using concepts of image classification , object detection, video processing.



2. Computer Vision

1. Object Detection using Google lens app can help you to identify the flower name in your garden.
2. Self driving cars like tesla, waymo to recognize lamp posts ,parking zones and identify difference between pedestrians and stop signals.
3. Face Recognition to lock unlock phones to hence security.
4. DeepFace system of Facebook to tag images.



3. Natural Language Processing (NLP)

- Natural Language Processing (NLP) refers to AI method of communicating with an intelligent systems using a natural language such as English
- It is the sub-field of AI that is focused on enabling computers to understand and process human language
- The ultimate objective of NLP is to read, decipher, understand, and make sense of the human languages in a manner that is valuable
- Most NLP techniques rely on machine learning to derive meaning from human languages
- Enables machines to understand, interpret, and generate human language, making communication with computers more natural

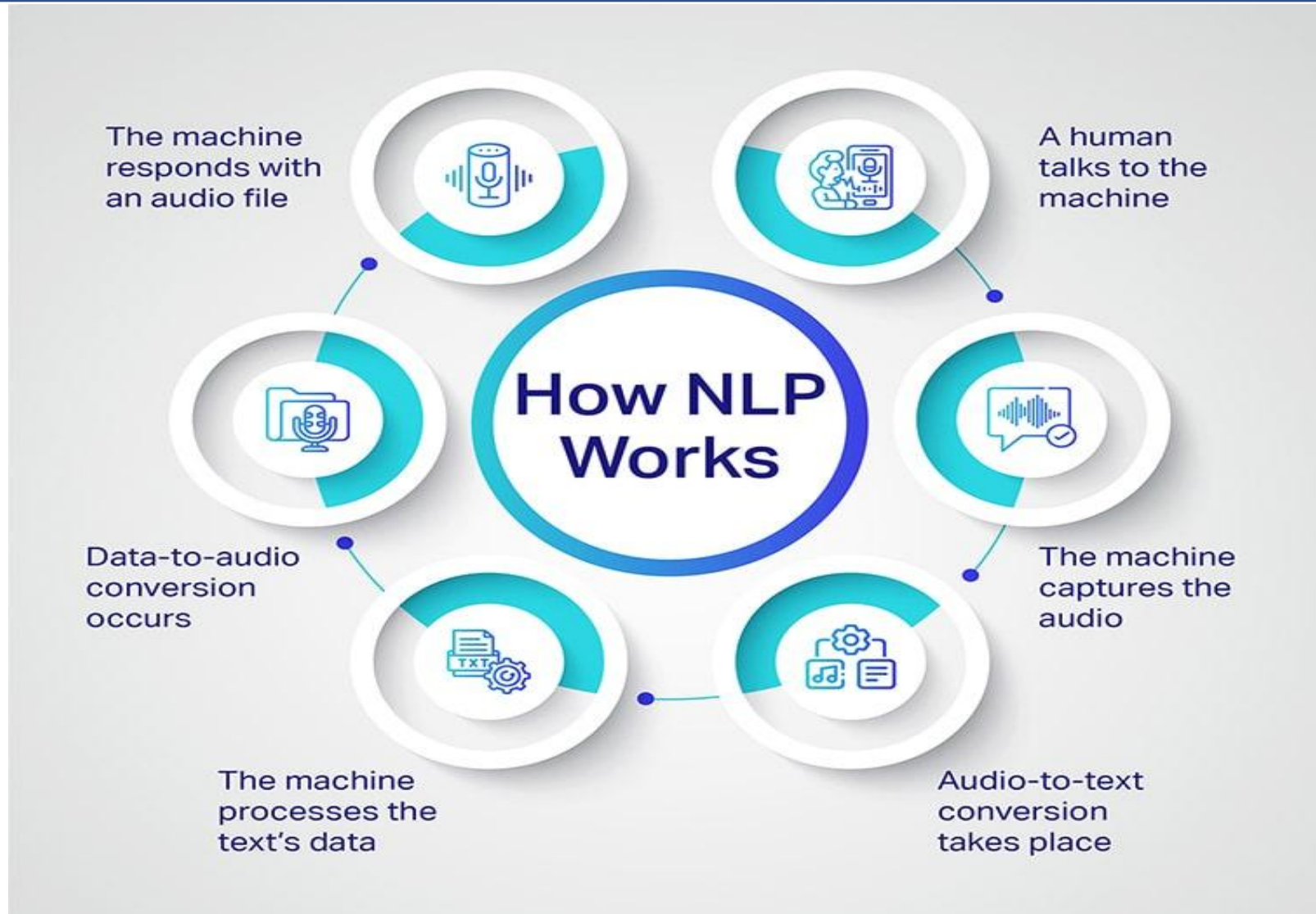


3. Natural Language Processing (NLP)

- There are two components of Natural Language Processing:
 1. Natural Language Understanding-
 - It helps machine to recognize natural speech and understand its meaning.
 - It is a process of reading and interpreting language.
 2. Natural Language Generation-
 - It helps in producing the response in natural language.
 - It is the process of writing or generating language.



How NLP Works? (like Alexa)



Steps in NLP

Lexical Analysis

It involves identifying and analyzing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of text into paragraphs, sentences, and words

Syntactic Analysis

It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words. The sentence such as “The school goes to boy” is rejected by English syntactic analyzer

Semantic Analysis

It draws the exact meaning or the dictionary meaning from the text. The text is checked for meaningfulness. It is done by mapping syntactic structures and objects in the task domain.

Discourse Integration

The meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.

Pragmatic Analysis

During this, what was said is re-interpreted on what it actually meant. It involves deriving those aspects of language which require real world knowledge



Uses of Natural Language Processing (NLP)



Amazon's Alexa and Apple's Siri are examples of intelligent **voice driven interfaces** that use NLP to respond to vocal prompts.

Twitter uses NLP to identify and filter out terrorist language in their tweets.



Uses of Natural Language Processing (NLP)

- NLP enables the recognition and **prediction of diseases** based on electronic health records and patient's own speech. NLP is particularly booming in the **healthcare industry**
- Organizations can determine what customers are saying about a service or product by identifying and extracting information in sources like social media (**sentiment analysis**)
- Companies like Yahoo and Google filter and classify your emails with NLP by analyzing text in emails that flow through their servers and **stopping spam** before they even enter your inbox
- To help **identifying fake news**, a system can be developed to determine if a source is accurate or politically biased, detecting if a news source can be trusted or not
- NLP is also being used in both the search and selection phases of **talent recruitment**
- NLP as Plagiarism checker



Thank You!!

